

Effect Heterogeneity Handout

This is my tau sub i. There are many like it, but this one is mine.

In causal inference, each unit has two potential outcomes: $Y_i(1)$ if treated and $Y_i(0)$ if not. At best, we can observe one of these, leading to the fundamental challenge of causal inference: moving from a correlational understanding (units differ in their average outcome when $W = 1$ versus $W = 0$) towards a causal statement of *individual* treatment effect: $\tau_i = Y_i(1) - Y_i(0)$.

In real-world settings, the effect can vary across individuals—known as effect heterogeneity. Formally, a Conditional Average Treatment Effect (CATE) is $\tau(x) = E[Y_i(1) - Y_i(0) | X = x]$. As the ATE is merely an aggregation of CATEs, identifying which covariates modify $\tau(x)$ can be crucial!

To estimate from data, we assume conditional ignorability (no unobserved confounding once we adjust for X) and overlap (non-zero probability of treatment/control for relevant subgroups). Under these assumptions, classic strategies include stratification by subgroups or interactions in regressions. However, when X is high dimensional or we are unsure of relevant subgroups, there are other options.

Modern machine learning helps uncover heterogeneity without enumerating all interactions. For example, *meta-learners* (Künzel et al. 2019) use flexible regressors to estimate. These approaches can reveal nuanced effect modifiers without prespecification, but generally need larger samples and produce more complex outputs (Rudolph et al. 2023). Integrating such models into our research can help us rigorously explore subgroup-specific signals that theory might not catch.

Considerations:

1. ML methods often require more data to reliably estimate local treatment effects.
2. Variable importance, second-stage regressions can highlight which covariates drive effect differences. Substantive expertise remains vital.
3. Software packages for random forests and bayesian additive regression trees are highly capable (Cheung et al. 2025) and are most often implemented in R and/or Python.

Causal forests (as implemented in the R **grf** package) combine the principle of honesty— separating data into “splitting” and “estimation” subsamples—with an R-learner approach that residualizes both the outcome Y and the treatment W . In other words, for each unit, they subtract the portion of Y or W that is explained by the covariates X . As splitting uses one subset of data and effect estimation uses another, the method balances bias and variance; combined with the R-learner’s orthogonal (‘locally robust’) objective, this yields CATE estimates that are first-order insensitive to small errors in the outcome and treatment models, and that approach near-oracle performance as those models improve.

Overall, causal forests are a powerful framework for assessing effect heterogeneity where confounding or variation is a concern like in the social, medical, or policy fields. There are several difficulties: sample size requirements exceed linear models, boosting algorithms often (slightly) outperform, interpretation is evolving and not always straightforward, and covariates must be numerized (See Johanneman et al. 2021 for tips on this). Lastly, alternative ML approaches such as gradient boosting or BART perform competitively and sometimes achieve slightly better predictive performance.

From Residualization, to Trees, to Treatment Effects

- ❖ To create a causal forest, the **grf** package residualizes Y and W on covariates X before running a causal forest. This residualization-then-learning strategy, called the R-learner (Nie and Wager 2021), maintains quasi-oracle properties even under partial misspecification.
 - Quasi-oracle estimators approach the performance of an oracle estimator that knew the true data generating process in advance—thus minimizing bias in estimation.
- ❖ Each tree in a forest is grown on a subsample of the data, which is itself split again for honesty:
 - Train: I_1 is used only to select splits based on a random subselection of covariates.
 - Test: I_2 is used only to estimate outcomes (or treatment effects) within leaves determined by those splits.
 - By preventing the tree from using the same data for both splitting and outcome estimation, we reduce the bias that can arise when splits are overfitted.
- ❖ After a forest of B honest trees is grown, each covariate vector x of I_2 is pushed through every tree to find a terminal leaf. Observations e in the same leaf as x in tree b are neighbors of x .
- ❖ If the leaf in tree b has $|Nb|$ observations, each neighbor e in that leaf contributes a weight of $1/|Nb|$. Summing across all B trees,

$$\alpha[e] = \frac{1}{B} \sum_{b=1}^B \frac{1\{e \in \text{leaf}_b(x)\}}{|Nb|}.$$

The resulting $\alpha[e]$ measures how often e co-occurs in the same leaf as x .

- ❖ To estimate the Conditional Average Treatment Effect (CATE) at x , the **grf** package solves a weighted objective where Y is the outcome, W is the treatment, across units i :

$$\hat{\tau}(x) = \arg \min(\theta) \sum_i \alpha[i] \text{Loss}(Y_i, W_i, \theta).$$

- ❖ Confidence intervals come from a bootstrapping-esque procedure that groups trees and then compares predictions within and across groups to estimate neighbor-weighted CATE stability.

FYI

1. Nearly jargon-free algorithm reference: grf-labs.github.io/grf/REFERENCE.html
2. Recent use of Causal Forests, featuring a meta-regression approach (Feb 11th 2025): Hiilamo, Aapo, Niina Metsä-Simola, Philipp Dierker, Pekka Martikainen, and Mikko Myrskylä. 2025. “Heterogenous Long-Term Health and Social Outcomes of Type 1 Diabetes - A Full Population 30-Year Observational Cohort Study.” *American Journal of Epidemiology* kwaf028. doi: [10.1093/aje/kwaf028](https://doi.org/10.1093/aje/kwaf028)
3. Athey, Susan, Julie Tibshirani, and Stefan Wager. 2019. “Generalized Random Forests.” *The Annals of Statistics* 47(2):1148–78. doi: [10.1214/18-AOS1709](https://doi.org/10.1214/18-AOS1709)
4. Nie, X., and S. Wager. 2021. “Quasi-Oracle Estimation of Heterogeneous Treatment Effects.” *Biometrika* 108(2):299–319. doi: [10.1093/biomet/asaa076](https://doi.org/10.1093/biomet/asaa076)

FURTHER FYI

5. Cheung, Michael, Anna Dimitrova, and Tarik Benmarhnia. 2025. “An Overview of Modern Machine Learning Methods for Effect Measure Modification Analyses in High-Dimensional Settings.” *SSM - Population Health* 29:101764. doi: [10.1016/j.ssmph.2025.101764](https://doi.org/10.1016/j.ssmph.2025.101764).
6. Johannemann, Jonathan, Vitor Hadad, Susan Athey, and Stefan Wager. 2021. “Sufficient Representations for Categorical Variables.” *arXiv* doi: [10.48550/arXiv.1908.09874](https://doi.org/10.48550/arXiv.1908.09874)
7. Künzel, Sören R., Jasjeet S. Sekhon, Peter J. Bickel, and Bin Yu. 2019. “Metalearners for Estimating Heterogeneous Treatment Effects Using Machine Learning.” *Proceedings of the National Academy of Sciences* 116(10):4156–65. doi: [10.1073/pnas.1804597116](https://doi.org/10.1073/pnas.1804597116).
8. Rudolph, Kara E., Nicholas T. Williams, Caleb H. Miles, Joseph Antonelli, and Ivan Diaz. 2023. “All Models Are Wrong, but Which Are Useful? Comparing Parametric and Nonparametric Estimation of Causal Effects in Finite Samples.” *Journal of Causal Inference* 11(1). doi: [10.1515/jci-2023-0022](https://doi.org/10.1515/jci-2023-0022).